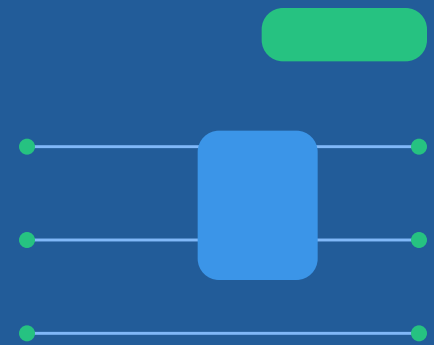


INPUT()

Function in Python

Day 19/55 - Make programs interactive

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First interactive idea

Ask the user, then use the answer.

`input()` pauses the program, waits for text, and returns that text to your code.

Video Summary

This Telusko tutorial explains how Python programs collect values with `input()`, why every input starts as a string, how `int()` fixes numeric addition, and how slicing can keep only the first character when the logic needs it.

1

Ask

Use `input()` to collect data while the program runs.

2

Convert

Numbers typed by users must be converted before math.

3

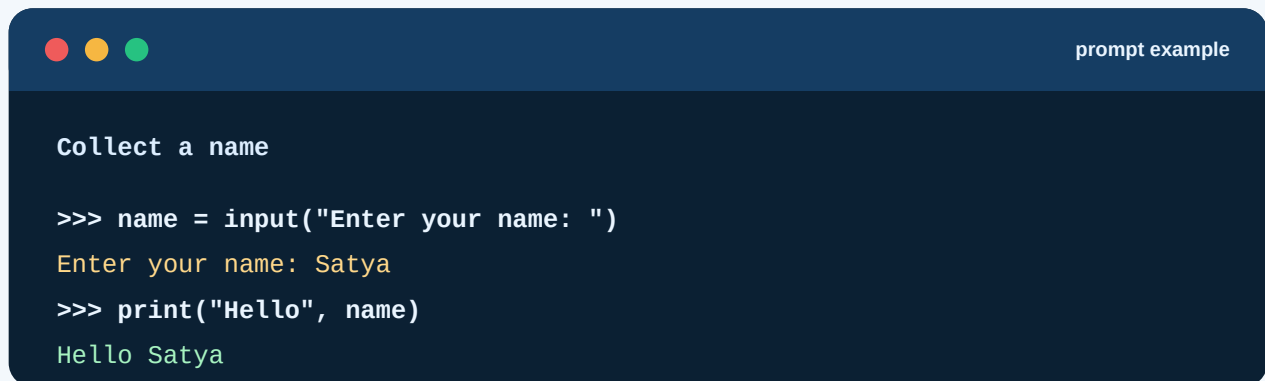
Slice

Use indexing when you only need one part of the answer.

1

Basic Usage of input()

The `input()` function pauses the program and waits for the user to type something. A prompt string inside `input()` tells the user what to enter.



```
Collect a name

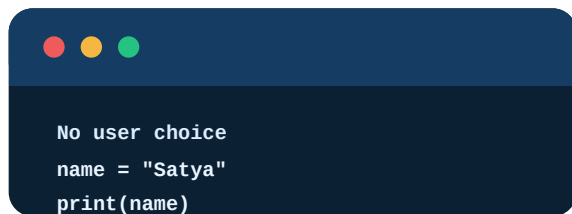
>>> name = input("Enter your name: ")
Enter your name: Satya
>>> print("Hello", name)
Hello Satya
```

2

From Fixed Values to Interactive Code

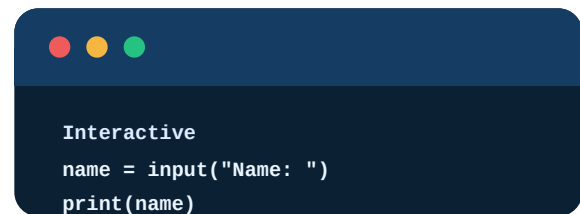
Hard-coded values are useful for practice. `input()` lets the same program work with different answers every time it runs.

Before: fixed value



```
No user choice
name = "Satya"
print(name)
```

After: user decides



```
Interactive
name = input("Name: ")
print(name)
```

1

Prompt

Python shows a question or label.



2

User Types

The answer comes from the keyboard.



3

Store

Assign the result to a variable.

3

input() Always Returns a String

Even if the user types digits, Python receives text first. That is why numeric input must be converted before arithmetic.

Common beginner trap

`'5' + '7'`

`'57'`

Text plus text joins the strings together.

Correct math path

`int("5") + int("7")`

`12`

Convert text digits into numbers first.

4

Type Conversion for Numeric Input

Wrap `input()` with `int()` when the answer should be a whole number. Use `float()` when decimals are allowed.

```
conversion example

Add two numbers

>>> a = int(input("Enter first number: "))
Enter first number: 5
>>> b = int(input("Enter second number: "))
Enter second number: 7
>>> a + b
12
```

User types

`"5"``input()`

string

`int()`

number



math

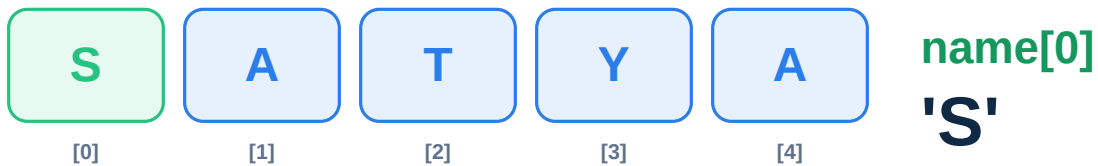
`5 + 7 = 12`

5

Data Handling and Slicing

Sometimes your program does not need the full answer. Python indexing lets you keep only the first character with `[0]`.

Example: capture one character



Only keep the first character

```
>>> name = input("Enter your name: ")
Enter your name: Satya
>>> first = name[0]
>>> first
'S'
```

Day 19/55 Key Takeaways

- `input()` makes a program interactive by asking the user for data.
- A prompt string tells the user what the program expects.
- `input()` returns text, so convert with `int()` or `float()` before math.
- Use `[0]` when the logic only needs the first character.

Source video: https://www.youtube.com/watch?v=Ye3vV4QQ_js&list=PLsyeobzWxl7omDoEYrrf3oXvXxa6MPgek&index=18